Dual-Damping Linearized ALM for Nonsmooth Composite Optimization

Luxuan Li[1]

[1]Beihang University

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Outline

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I. Problem Setting

Structured Nonsmooth Composite Optimization

We consider

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \quad F(\boldsymbol{x}) := f(\boldsymbol{x}) + h\big(\mathcal{A}(\boldsymbol{x})\big). \tag{P0}$$

- $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and may be nonconvex.
- $\mathcal{A} : \mathbb{R}^n \to \mathbb{R}^m$ is continuously differentiable and may be nonlinear.
- $h : \mathbb{R}^m \to \mathbb{R}$ is convex, Lipschitz continuous, and possibly nonsmooth.

Approximate KKT Point

- Introduce $\boldsymbol{y} = \mathcal{A}(\boldsymbol{x})$:

$$\min_{\boldsymbol{x},\boldsymbol{y}} \quad f(\boldsymbol{x}) + h(\boldsymbol{y}) \qquad \text{s.t.} \qquad \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} = \boldsymbol{0}. \tag{P}$$

- The split KKT system is

$$\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \boldsymbol{z} = \boldsymbol{0}, \qquad \boldsymbol{z} \in \partial h(\boldsymbol{y}), \qquad \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} = \boldsymbol{0}.$$

- **approximate ε-KKT point.** Find $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z})$ such that

$$\|\nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \boldsymbol{z}\| \le \varepsilon, \quad \boldsymbol{z} \in \partial h(\boldsymbol{y}), \quad \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\| \le \varepsilon.$$

- The exact $\boldsymbol{y}$-proximal step makes the subdifferential condition exact.

Main Results

Method	**Constrained setting**	A	**Nonsmoothness of h**	**Loop**	KKT measure	Penalty ρ	Dual regularizer s	**Iteration complexity**
MRSALM	Manifold	Linear	Lipschitz	✓	ε-KKT	fixed	$s_k = \Theta(k^{1/4}) \uparrow$	$O(\varepsilon^{-4})$
MRALM	Manifold	Linear	Lipschitz	×	ε-KKT	fixed	$s_k = \Theta(k^{1/3}) \uparrow$	$O(\varepsilon^{-3})$
DDLALM (Ours)	Uncon-strained	Non-linear	Lipschitz	✓	ε-KKT	fixed	$s_k = \Theta(k^{1/3}) \uparrow$	$O(\varepsilon^{-3})$

II. Dual-Damping Reformulation

Dual-Proximal Augmented Lagrangian

- The Augmented Lagrangian of Problem (P) is defined as

$$\mathcal{L}_\rho(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle \boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} \rangle + \frac{\rho}{2} \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2.$$

- Dual-regularized Augmented Lagrangian: Fix a center multiplier $\boldsymbol{\mu}$ and introduce $s > 0$, we define

$$\mathcal{P}_s(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}) := \mathcal{L}_\rho(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}) - \frac{1}{2s} \|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2.$$

- For fixed $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\mu})$, $\boldsymbol{\lambda} \mapsto \mathcal{P}_s$ is $1/s$-strongly concave, admitting a unique maximizer

$$\boldsymbol{\lambda}_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}) = \boldsymbol{\mu} + s\big(\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\big).$$

- nonconvex–concave $\min_{x,y} \max_\lambda \mathcal{L}_\rho(x, y, \lambda) \longrightarrow$ nonconvex–strongly concave problem:

$$\min_{\boldsymbol{x}, \boldsymbol{y}} \max_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}).$$

- **Reduced Augmented Lagrangian.** Substitution $\boldsymbol{\lambda}_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu})$ into $\mathcal{P}_s$ yields the value function:

$$\begin{aligned}\Phi_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}) &:= \max_{\boldsymbol{\lambda}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}) \\ &= f(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle \boldsymbol{\mu}, \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} \rangle + \frac{\rho + s}{2} \|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2.\end{aligned}$$

Danskin's Theorem for the Value Function $\Phi_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu})$

Danskin's theorem gives the smooth outer gradients: ▸ Theorem →

- **$\boldsymbol{x}$-gradient:**

$$\begin{aligned}\nabla_{\boldsymbol{x}} \Phi_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}) &= \nabla_{\boldsymbol{x}} \mathcal{P}_s\big(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}); \boldsymbol{\mu}\big) \\ &= \nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \left[\boldsymbol{\mu} + (\rho + s)\big(\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\big)\right].\end{aligned}$$

- **$\boldsymbol{y}$-subgradient:**

$$\partial_{\boldsymbol{y}} \Phi_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}) = \partial h(\boldsymbol{y}) - \boldsymbol{\mu} - (\rho + s)\big(\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\big).$$

- **$\boldsymbol{\mu}$-gradient:**

$$\begin{aligned}\nabla_{\boldsymbol{\mu}} \Phi_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}) &= \nabla_{\boldsymbol{\mu}} \mathcal{P}_s\big(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}); \boldsymbol{\mu}\big) \\ &= \frac{1}{s}\big(\boldsymbol{\lambda}_s(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\mu}) - \boldsymbol{\mu}\big) = \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}.\end{aligned}$$

III. Dual-Damping Linearized ALM

Core Framework: Coupled Saddle Update

Idea: Apply gradient steps on Φ_{s_k} and solve the coupled $(\boldsymbol{y}, \boldsymbol{\lambda})$ saddle block exactly.
Parameter: tolerance $\varepsilon > 0$, η_k, τ_k, s_k, max iteration number K.
Set $\boldsymbol{r}^k := \mathcal{A}(\boldsymbol{x}^k) - \boldsymbol{y}^k$ and $\rho_{s_k} := \rho + s_k$.

- **Steps 1–2: x-gradient descent.**

$$\begin{aligned} \boldsymbol{x}^{k+1} &= \boldsymbol{x}^k - \eta_k \nabla_{\boldsymbol{x}} \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \\ &= \boldsymbol{x}^k - \eta_k \left[\nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top \left(\boldsymbol{\mu}^k + \rho_{s_k} \boldsymbol{r}^k \right) \right]. \end{aligned} \tag{1}$$

- **Step 3: coupled (y, λ) saddle update.**

$$(\boldsymbol{y}^{k+1}, \boldsymbol{\lambda}^{k+1}) = \operatorname{saddle}_{\boldsymbol{y},\boldsymbol{\lambda}} \mathcal{P}_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}^k). \tag{2}$$

- **Step 4: μ-gradient ascent on Φ_{s_k}.**

$$\begin{aligned} \boldsymbol{\mu}^{k+1} &= \boldsymbol{\mu}^k + \tau_k \nabla_{\boldsymbol{\mu}} \Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^k) \\ &= \boldsymbol{\mu}^k + \tau_k \boldsymbol{r}^{k+1}. \end{aligned} \tag{3}$$

Step 3: Coupled $(\boldsymbol{y}, \boldsymbol{\lambda})$ Saddle Update

With $(\boldsymbol{x}^{k+1}, \boldsymbol{\mu}^k)$ fixed, solve the saddle subproblem exactly:

$$\min_{\boldsymbol{y}} \max_{\boldsymbol{\lambda}} \ \mathcal{P}_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}^k).$$

Step 3: Executable form: ▸ Why $\rho + s_k$?

$$\boldsymbol{y}^{k+1} = \operatorname{prox}_{h/\rho s_k}\left(\mathcal{A}(\boldsymbol{x}^{k+1}) + \frac{\boldsymbol{\mu}^k}{\rho s_k}\right),$$
$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\mu}^k + s_k \boldsymbol{r}^{k+1}.$$

▷ **Coupled saddle point:**

$$\boldsymbol{y}^{k+1} = \arg\min_{\boldsymbol{y}} \ \mathcal{P}_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}, \boldsymbol{\lambda}^{k+1}; \boldsymbol{\mu}^k) = \operatorname{prox}_{h/\rho}\left(\mathcal{A}(\boldsymbol{x}^{k+1}) + \frac{\boldsymbol{\lambda}^{k+1}}{\rho}\right),$$
$$\boldsymbol{\lambda}^{k+1} = \arg\max_{\boldsymbol{\lambda}} \ \mathcal{P}_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}, \boldsymbol{\lambda}; \boldsymbol{\mu}^k) = \boldsymbol{\mu}^k + s_k\big(\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}^{k+1}\big).$$

▷ **Optimality conditions:**

$$0 \in \partial h(\boldsymbol{y}^{k+1}) - \boldsymbol{\lambda}^{k+1} + \rho\big(\boldsymbol{y}^{k+1} - \mathcal{A}(\boldsymbol{x}^{k+1})\big),$$
$$0 = \mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}^{k+1} - \frac{1}{s_k}(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k).$$

Key Identities Used Throughout

- **Exact identities**:

$$\begin{aligned}\boldsymbol{\lambda}^{k+1} &= \boldsymbol{\mu}^k + s_k \boldsymbol{r}^{k+1},\\ \boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k &= \tau_k \boldsymbol{r}^{k+1} = \frac{\tau_k}{s_k}(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k),\\ \boldsymbol{\lambda}^{k+1} + \rho \boldsymbol{r}^{k+1} &= \boldsymbol{\mu}^k + \rho_{s_k} \boldsymbol{r}^{k+1} \in \partial h(\boldsymbol{y}^{k+1}).\end{aligned} \tag{4}$$

- **Convex-combination identities**

$$\begin{aligned}\boldsymbol{\lambda}^{k+1} &= \frac{\rho}{\rho_{s_k}} \boldsymbol{\mu}^k + \frac{s_k}{\rho_{s_k}}(\boldsymbol{\lambda}^{k+1} + \rho \boldsymbol{r}^{k+1}),\\ \boldsymbol{\mu}^{k+1} &= \left(1 - \frac{\tau_k}{s_k}\right) \boldsymbol{\mu}^k + \frac{\tau_k}{s_k} \boldsymbol{\lambda}^{k+1}.\end{aligned} \tag{5}$$

IV. Complexity Analysis

Convergence: $O(\varepsilon^{-3})$ Iteration Complexity

Under mild assumptions, choose

$$s_k = c_s(k+1)^{1/3}, \quad \eta_k = \frac{c_x}{s_k}, \quad \tau_k = \frac{\theta_\mu s_k}{k+2} = \Theta(k^{-2/3}). \quad c_x \widehat{L}\left(1 + \frac{\rho}{s_0}\right) \leq \frac{1}{2}.$$

where c_s, c_x, θ_μ, ρ, and $\widehat{L}$ are positive constants independent of k. Then there exists an index $R \in \{\lceil K/2 \rceil, \ldots, K-1\}$ such that $\boldsymbol{\lambda}^R + \rho \boldsymbol{r}^R \in \partial h(\boldsymbol{y}^R)$ and

$$\left\| \nabla f(\boldsymbol{x}^R) + J_{\mathcal{A}}(\boldsymbol{x}^R)^\top \left(\boldsymbol{\lambda}^R + \rho \boldsymbol{r}^R\right) \right\| = O\left(K^{-1/3}\right), \qquad \|\mathcal{A}(\boldsymbol{x}^R) - \boldsymbol{y}^R\| = O(K^{-1/3}).$$

Consequently, the oracle complexity of DDLALM to reach an approximate ε-KKT is of the order $\mathcal{O}(\varepsilon^{-3})$.

Main Assumptions

(i) **Level-bounded objective.** For every $\alpha \in \mathbb{R}$, the set

$$\mathcal{S}_\alpha := \{\boldsymbol{x} \in \mathbb{R}^n : F(\boldsymbol{x}) \leq \alpha\}$$

is empty or compact.

(ii) **Smoothness and boundedness**[1]**.** Given a nonempty working level set $\mathcal{S} := \mathcal{S}_{\alpha_0}$. The functions f and $\mathcal{A}$ are L_f- and $L_{\mathcal{A}}$-smooth on $\mathcal{S}$. Moreover,

$$\|\nabla f(\boldsymbol{x})\| \leq G_f, \quad \|J_{\mathcal{A}}(\boldsymbol{x})\| \leq G_{\mathcal{A}}, \quad M_{\mathcal{A}} := \sup_{\boldsymbol{x}\in\mathcal{S}} \|\mathcal{A}(\boldsymbol{x})\| < \infty.$$

(iii) h is proper, closed, convex, and globally L_h-Lipschitz; equivalently,

$$\|\boldsymbol{z}\| \leq L_h \qquad \text{for all } \boldsymbol{z} \in \partial h(\boldsymbol{y}).$$

Remark

The smoothness condition can be imposed on a compact convex set $\mathcal{S}$ satisfying $\mathcal{S}_\alpha \subseteq \operatorname{int}(\mathcal{S})$.

The containment $\boldsymbol{x}^k \in \mathcal{S}_\alpha$ for every $k \geq 0$ is established by induction. ▸ Joint induction claim →

[1] Y. Xie and S. J. Wright, "Complexity of Proximal Augmented Lagrangian for Nonconvex Optimization with Nonlinear Equality Constraints," *J. Sci. Comput.*, 86:38, 2021.

Proof Roadmap

- **Step 1: establish multiplier boundedness and feasibility control.**
  Assumption (iii) provides bounded subgradients of h. The two convex-combination identities in (5) imply

$$\|\boldsymbol{\lambda}^{k+1}\|,\ \|\boldsymbol{\mu}^{k+1}\| \le M_\mu, \qquad \|\boldsymbol{r}^{k+1}\| \le \frac{C_r}{\rho_{s_k}}.$$

- **Step 2: sufficient primal descent on the value function Φ_s.**
  The $\boldsymbol{x}$-gradient step and the exact $(\boldsymbol{y}, \boldsymbol{\lambda})$ saddle solve in in (2) yield sufficient primal descent; Exactness eliminates the multiplier-tracking error with respect to $\boldsymbol{\lambda}$. Together, these properties yield

$$\Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^k) - \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \le -\frac{3\rho_{s_k}}{4\theta_x}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{\rho_{s_k}}{2}\|\Delta_{\boldsymbol{y}}^{k+1}\|^2.$$

- **Step 3: control the moving-potential drift.** Using $\Delta_{\boldsymbol{\mu}}^{k+1} = \tau_k \boldsymbol{r}^{k+1}$ from (4), the two shifts (s_k and $\boldsymbol{\mu}^k$) satisfy

$$\Phi_{s_{k+1}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) - \Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^k) = \underbrace{\langle \Delta_{\boldsymbol{\mu}}^{k+1}, \boldsymbol{r}^{k+1}\rangle + \frac{\rho_{s_{k+1}} - \rho_{s_k}}{2}\|\boldsymbol{r}^{k+1}\|^2}_{\text{moving-potential drift}}$$

$$\le \underbrace{C_r^2\left(\frac{\tau_k}{\rho_{s_k}^2} + \frac{\rho_{s_{k+1}} - \rho_{s_k}}{2\rho_{s_k}^2}\right)}_{e_k}$$

- **Step 4: choose the parameter schedules and control the KKT residuals.**
  The parameter schedules make the drift errors summable:

$$\sum_{k=0}^{\infty} e_k < \infty$$

▸ Appendix I: Lemmas 1–4 →

Comparison with Closely Related Methods

Method	Problem setting	Convexity of h	KKT measure	Penalty ρ	Parameter schedule	Iteration complexity
SDCAM$_{1\ell}$ [1]	Composite	✓	$(\varepsilon_1, \varepsilon_2, 0)$ stationarity	N/A	Continuation ↑	$O(\varepsilon^{-3})$
SDD-ADMM [2]	Nonlinear equality	✓(ncvx)	ε-stationarity	$\Theta(\varepsilon^{-1})$	ω, τ fixed per run	$O(\varepsilon^{-3})$
DDLALM [This work]	$h(\mathcal{A}(\boldsymbol{x}))$	✓	**split $\boldsymbol{\varepsilon}$-KKT**	**fixed**	$s_k = \Theta(k^{1/3})$ ↑	$O(\varepsilon^{-3})$

SDCAM$_{1\ell}$ is the closest same-scope complexity comparator; SDD-ADMM is the nonlinear ALM/ADMM-family comparator.

V. Discussion

Concern I: Can the y-Update Avoid s_k in DDALM?

Recalling

$$\mathcal{P}_{s_k}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}) := f(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle \boldsymbol{\lambda}, \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y} \rangle + \frac{\rho}{2}\|\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}\|^2 - \frac{1}{2s_k}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2.$$

For fixed $(\boldsymbol{x}, \boldsymbol{\mu})$, $\mathcal{P}_{s_k}$ is ρ-strongly convex in $\boldsymbol{y}$ and $1/s_k$-strongly concave in $\boldsymbol{\lambda}$. Hence

$$\min_{\boldsymbol{y}} \max_{\boldsymbol{\lambda}} \mathcal{P}_{s_k} = \max_{\boldsymbol{\lambda}} \min_{\boldsymbol{y}} \mathcal{P}_{s_k},$$

Both problems attain the same unique saddle point $(\boldsymbol{y}^\star, \boldsymbol{\lambda}^\star)$: ▸ Why $\rho + s_k$?

$$\boldsymbol{y}^\star = \operatorname{prox}_{h/(\rho+s_k)}\left(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\mu}}{\rho + s_k}\right),$$
$$\boldsymbol{\lambda}^\star = \boldsymbol{\mu} + s_k\left(\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}^\star\right).$$

Two different one-step update strategies:

- DDLALM: solve the saddle problem **exactly.** The answer seems to be not!
- Candidate route: solve the saddle problem **inexactly.**

Candidate Solution: Reduced-Dual Route

- **Eliminate the y-block through the ρ-prox:**

$$d_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{\mu}) := \min_{\boldsymbol{y}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}), \qquad \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}) = \operatorname{prox}_{h/\rho}\left(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\lambda}}{\rho}\right).$$

- **Obtain an exact subdifferential certificate:**

$$\boldsymbol{\lambda} + \rho\big[\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x})\big] \in \partial h\big(\boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x})\big).$$

- **Apply Danskin's theorem:**

▸ Danskin derivation →

$$\nabla_{\boldsymbol{x}} d_s = \nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \left\{\boldsymbol{\lambda} + \rho\big[\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x})\big]\right\},$$

$$\nabla_{\boldsymbol{\lambda}} d_s = \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}) - \frac{\boldsymbol{\lambda} - \boldsymbol{\mu}}{s}, \qquad \nabla_{\boldsymbol{\mu}} d_s = \frac{\boldsymbol{\lambda} - \boldsymbol{\mu}}{s}.$$

Moving-center identity

$$\boxed{\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}) = \nabla_{\boldsymbol{\lambda}} d_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{\mu}) + \nabla_{\boldsymbol{\mu}} d_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{\mu})}$$

The $\boldsymbol{\mu}$-gradient exactly complements the regularization term in the $\boldsymbol{\lambda}$-gradient.

Candidate Algorithm

Idea: Apply $\boldsymbol{x}$-descent and $(\boldsymbol{\lambda}, \boldsymbol{\mu})$-ascent steps to the value function d_{s_k}.
Parameters: tolerance $\varepsilon > 0$, stepsizes η_k, α_k, τ_k, regularization s_k, and maximum iteration count K.

- **Use the same ρ-prox mapping throughout:**

$$\boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}) = \operatorname{prox}_{h/\rho}\left(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\lambda}}{\rho}\right).$$

- **$\boldsymbol{x}$-gradient descent on $d_{s_k}(\cdot, \boldsymbol{\lambda}^k; \boldsymbol{\mu}^k)$:**

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \eta_k \left\{\nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top \left[\boldsymbol{\lambda}^k + \rho\big(\mathcal{A}(\boldsymbol{x}^k) - \boldsymbol{y}(\boldsymbol{\lambda}^k; \boldsymbol{x}^k)\big)\right]\right\}.$$

- **One reduced-dual $\boldsymbol{\lambda}$-gradient ascent step:**

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\lambda}^k + \alpha_k \left[\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}(\boldsymbol{\lambda}^k; \boldsymbol{x}^{k+1}) - \frac{\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k}{s_k}\right].$$

- **Moving-center $\boldsymbol{\mu}$-gradient ascent:**

$$\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \tau_k \nabla_{\boldsymbol{\mu}} d_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}^{k+1}; \boldsymbol{\mu}^k) = \boldsymbol{\mu}^k + \frac{\tau_k}{s_k}(\boldsymbol{\lambda}^{k+1} - \boldsymbol{\mu}^k), \qquad 0 < \tau_k < s_k.$$

Oracle cost

The $\boldsymbol{x}$-first order evaluates the same ρ-prox mapping at $(\boldsymbol{\lambda}^k, \boldsymbol{x}^k)$ and $(\boldsymbol{\lambda}^k, \boldsymbol{x}^{k+1})$.

Exact $\boldsymbol{\lambda}$-Block Solve as a Special Case

- **Choose the reduced-dual stepsize $\alpha_k = s_k$:**

$$\begin{aligned}\boldsymbol{\lambda}^{k+1} &= \boldsymbol{\lambda}^k + s_k \left[\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}(\boldsymbol{\lambda}^k; \boldsymbol{x}^{k+1}) - \frac{\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k}{s_k}\right] \\ &= \boldsymbol{\mu}^k + s_k \left[\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}(\boldsymbol{\lambda}^k; \boldsymbol{x}^{k+1})\right].\end{aligned}$$

- **This is an exact solve of the frozen-$\boldsymbol{y}$ multiplier block:**

$$\boldsymbol{\lambda}^{k+1} = \arg\max_{\boldsymbol{\lambda}} \mathcal{P}_{s_k}\left(\boldsymbol{x}^{k+1}, \boldsymbol{y}(\boldsymbol{\lambda}^k; \boldsymbol{x}^{k+1}), \boldsymbol{\lambda}; \boldsymbol{\mu}^k\right).$$

- **The distinction from reduced-dual exact maximization:** in $d_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{\lambda}; \boldsymbol{\mu}^k)$, the minimizer $\boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}^{k+1})$ changes with $\boldsymbol{\lambda}$.

Interpretation

$\alpha_k = s_k$ is exact for the frozen-$\boldsymbol{y}$ block, but it is not an exact maximizer of d_{s_k}.

The role of α_k

- **Core bottleneck:**
  primal descent $\leq$ $-$descent $+ C\eta_k \|\lambda^k - \lambda_k^\star(x^k)\|^2 +$ regularization drift, which limits the η_k.
- **The frozen-y exact step is not the stable reduced-dual scale:**

$$L_{\lambda,k} \leq \frac{1}{\rho} + \frac{1}{s_k}, \qquad \alpha_k \leq \frac{1}{\frac{1}{\rho} + \frac{1}{s_k}} = \frac{\rho s_k}{\rho + s_k} \leq \frac{1}{L_{\lambda,k}}.$$

  Thus $\alpha_k = s_k$ does not respect this scale when $s_k \uparrow$.
- **Proposed stable dual step:**

$$\alpha_k = \theta_\lambda \frac{\rho s_k}{\rho + s_k}, \qquad 0 < \theta_\lambda \leq 1.$$

  It yields the convex-combination representation

$$\boldsymbol{\lambda}^{k+1} = (1-\theta_\lambda)\boldsymbol{\lambda}^k + \theta_\lambda \frac{s_k}{\rho + s_k} \left\{ \boldsymbol{\lambda}^k + \rho\left[\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}(\boldsymbol{\lambda}^k; \boldsymbol{x}^{k+1})\right] \right\} + \theta_\lambda \frac{\rho}{\rho + s_k} \boldsymbol{\mu}^k.$$

- **Main parameter schedule:**

$$s_k = c_s (k+1)^a, \qquad \eta_k = \frac{\theta_x}{\rho + s_k}, \qquad \alpha_k = \theta_\lambda \frac{\rho s_k}{\rho + s_k}, \qquad \tau_k = \theta_\mu s_k,$$

  where $a \in (0,1)$, $c_s > 0$, $0 < \theta_\lambda \leq 1$, and $0 < \theta_\mu < 1$. The complexity target uses $a = 1/3$.

Question

Can stable dual contraction and constant-rate center tracking jointly absorb the multiplier tracking error?

References

[1] H. Zhang, N. Marumo, T. K. Pong, and A. Takeda, "Complexity and Convergence Analysis of a Single-Loop SDCAM for Lipschitz Composite Optimization and Beyond," arXiv:2512.24059, 2025.

[2] K. Sun and X. A. Sun, "Dual Descent Augmented Lagrangian Method and Alternating Direction Method of Multipliers," *SIAM Journal on Optimization*, 34(2):1679–1707, 2024.

[3] J. M. Danskin, "The Theory of Max–Min, with Applications," *SIAM Journal on Applied Mathematics*, 14(4):641–664, 1966.

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Appendix I: Lemma 1—Multiplier and Feasibility Bounds

Contents

Suppose $0 < \tau_k \le s_k$. Define

$$M_\mu := \max\{\|\boldsymbol{\mu}^0\|, L_h\}, \qquad C_r := L_h + M_\mu.$$

Conclusion: For every $k \ge 0$,

$$\|\boldsymbol{\lambda}^{k+1}\| \le M_\mu, \qquad \|\boldsymbol{\mu}^{k+1}\| \le M_\mu, \tag{6}$$

$$\|\boldsymbol{r}^{k+1}\| \le \frac{C_r}{\rho_{s_k}}, \qquad \frac{\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\|^2}{\tau_k} \le \frac{C_r^2 \tau_k}{\rho_{s_k}^2}. \tag{7}$$

Proof:

- **Saddle certificate.** Assumption (iii) and the saddle certificate in (4) give

$$\boldsymbol{\lambda}^{k+1} + \rho \boldsymbol{r}^{k+1} \in \partial h(\boldsymbol{y}^{k+1}), \qquad \|\boldsymbol{\lambda}^{k+1} + \rho \boldsymbol{r}^{k+1}\| \le L_h.$$

- **Two-multiplier boundedness.** The base case holds since $\|\boldsymbol{\mu}^0\| \le M_\mu$. Assume $\|\boldsymbol{\mu}^k\| \le M_\mu$. Applying the two convex combinations in (5), respectively, gives

$$\|\boldsymbol{\lambda}^{k+1}\| \le \frac{\rho}{\rho_{s_k}} M_\mu + \frac{s_k}{\rho_{s_k}} L_h \le M_\mu,$$

$$\|\boldsymbol{\mu}^{k+1}\| \le \left(1 - \frac{\tau_k}{s_k}\right) M_\mu + \frac{\tau_k}{s_k} M_\mu = M_\mu.$$

  Thus, (6) follows by induction.

- **Feasibility and center-multiplier drift.** Since $\rho_{s_k} \boldsymbol{r}^{k+1} = (\boldsymbol{\lambda}^{k+1} + \rho \boldsymbol{r}^{k+1}) - \boldsymbol{\mu}^k$ and $\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k = \tau_k \boldsymbol{r}^{k+1}$ by (4),

$$\|\boldsymbol{r}^{k+1}\| \le \frac{L_h + M_\mu}{\rho_{s_k}} = \frac{C_r}{\rho_{s_k}},$$

$$\frac{\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\|^2}{\tau_k} = \tau_k \|\boldsymbol{r}^{k+1}\|^2 \le \frac{C_r^2 \tau_k}{\rho_{s_k}^2}.$$

Appendix I: Lemma 2—Quasi-Descent on Φ_s

Contents

Let $\Delta_{\boldsymbol{x}}^{k+1} := \boldsymbol{x}^{k+1} - \boldsymbol{x}^k$, and define $\Delta_{\boldsymbol{y}}^{k+1}$ and $\Delta_{\boldsymbol{\mu}}^{k+1}$ similarly.

Conclusion:

$$\begin{aligned}&\Phi_{s_{k+1}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) - \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \\ &\quad \le -\frac{3\rho_{s_k}}{4\theta_x}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2 - \frac{\rho_{s_k}}{2}\|\Delta_{\boldsymbol{y}}^{k+1}\|^2 + \left(\frac{1}{\tau_k} + \frac{\rho_{s_{k+1}} - \rho_{s_k}}{2\tau_k^2}\right)\|\Delta_{\boldsymbol{\mu}}^{k+1}\|^2.\end{aligned} \tag{8}$$

Proof:

- $\boldsymbol{x}$**-block descent.** With $\eta_k = \theta_x/\rho_{s_k}$, the $\boldsymbol{x}$-gradient step gives

$$\Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^k; \boldsymbol{\mu}^k) - \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \le -\frac{3\rho_{s_k}}{4\theta_x}\|\Delta_{\boldsymbol{x}}^{k+1}\|^2.$$

- $\boldsymbol{y}$**-block descent.** The exact $(\boldsymbol{y}, \boldsymbol{\lambda})$ saddle solve makes $\boldsymbol{y}^{k+1}$ the exact minimizer of the strongly convex $\boldsymbol{y}$-subproblem. Hence,

$$\Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^k) - \Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \le -\frac{\rho_{s_k}}{2}\|\Delta_{\boldsymbol{y}}^{k+1}\|^2.$$

- **Center-multiplier shift.** Using $\Delta_{\boldsymbol{\mu}}^{k+1} = \tau_k \boldsymbol{r}^{k+1}$ from (4),

$$\Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) - \Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^k) = \frac{1}{\tau_k}\|\Delta_{\boldsymbol{\mu}}^{k+1}\|^2.$$

- **Regularization shift.** The variation of s_k contributes

$$\Phi_{s_{k+1}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) - \Phi_{s_k}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) = \frac{\rho_{s_{k+1}} - \rho_{s_k}}{2\tau_k^2}\|\Delta_{\boldsymbol{\mu}}^{k+1}\|^2.$$

Appendix I: Lemma 3—KKT Residual Control

Contents

Fix $k \in \{\lceil K/2 \rceil, \ldots, K-1\}$. Then

$$\boldsymbol{\lambda}^k + \rho \boldsymbol{r}^k \in \partial h(\boldsymbol{y}^k),$$

$$\left\| \nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top (\boldsymbol{\lambda}^k + \rho \boldsymbol{r}^k) \right\| \leq \eta_k^{-1} \|\Delta_{\boldsymbol{x}}^{k+1}\| + G_{\mathcal{A}} C_r \left(\frac{\tau_{k-1}}{\rho_{s_{k-1}}} + \frac{\rho_{s_k} - \rho_{s_{k-1}}}{\rho_{s_{k-1}}} \right),$$

$$\|\mathcal{A}(\boldsymbol{x}^k) - \boldsymbol{y}^k\| \leq \frac{C_r}{\rho_{s_{k-1}}}$$

Proof:

- **Choose the KKT multiplier.** By (4) at iteration $k-1$,

$$\boldsymbol{\lambda}^k + \rho \boldsymbol{r}^k = \boldsymbol{\mu}^{k-1} + \rho_{s_{k-1}} \boldsymbol{r}^k \in \partial h(\boldsymbol{y}^k). \tag{9}$$

- **$\boldsymbol{x}$-step optimality.** The $\boldsymbol{x}$-update gives

$$\begin{aligned} \mathbf{0} &= \eta_k^{-1} \Delta_{\boldsymbol{x}}^{k+1} + \nabla_{\boldsymbol{x}} \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \\ &= \eta_k^{-1} \Delta_{\boldsymbol{x}}^{k+1} + \nabla f(\boldsymbol{x}^k) + J_{\mathcal{A}}(\boldsymbol{x}^k)^\top (\boldsymbol{\mu}^k + \rho_{s_k} \boldsymbol{r}^k). \end{aligned} \tag{10}$$

- **Multiplier mismatch.** Using $\boldsymbol{\mu}^k - \boldsymbol{\mu}^{k-1} = \tau_{k-1} \boldsymbol{r}^k$,

$$\begin{aligned} &\boldsymbol{\mu}^k + \rho_{s_k} \boldsymbol{r}^k - (\boldsymbol{\lambda}^k + \rho \boldsymbol{r}^k) \\ &\quad = (\tau_{k-1} + \rho_{s_k} - \rho_{s_{k-1}}) \boldsymbol{r}^k. \end{aligned}$$

Hence, (7) and $\|J_{\mathcal{A}}(\boldsymbol{x}^k)\| \leq G_{\mathcal{A}}$ give the results.

Appendix I: Lemma 4—Weighted Finite Length

Contents

Let $D_k := \|\Delta_{\boldsymbol{x}}^{k+1}\|^2 + \|\Delta_{\boldsymbol{y}}^{k+1}\|^2$ and $c_{\mathrm{desc}} := \min\{3/(4\theta_x), 1/2\} > 0$.
Condition: Let $s_k = c_s(k+1)^a$ and $\tau_k = \theta_\mu s_k/(k+2)$ with $a \in (0,1)$. Then

$$c_{\mathrm{desc}} \sum_{k=0}^{K-1} \rho_{s_k} \|\Delta_{\boldsymbol{x}}^{k+1}\|^2 \le c_{\mathrm{desc}} \sum_{k=0}^{K-1} \rho_{s_k} D_k \le C_{\mathrm{desc}} < \infty. \tag{11}$$

Proof:

- **The moving-potential error is summable.**

$$e_k := C_r^2 \left(\frac{\tau_k}{\rho_{s_k}^2} + \frac{\rho_{s_{k+1}} - \rho_{s_k}}{2\rho_{s_k}^2} \right),$$

$$\frac{\tau_k}{\rho_{s_k}^2} = O(k^{-(1+a)}), \qquad \frac{\rho_{s_{k+1}} - \rho_{s_k}}{\rho_{s_k}^2} = O(k^{-(1+a)}).$$

  Hence, $\sum_{k=0}^{\infty} e_k < \infty$.
- **Quasi-descent leaves a finite error budget.** Lemma 2 and Lemma 1 yield

$$\Phi_{s_{k+1}}^{k+1} - \Phi_{s_k}^k \le -c_{\mathrm{desc}} \rho_{s_k} D_k + e_k, \qquad \Phi_{s_k}^k := \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k). \tag{12}$$

- **Telescoping converts descent into a global budget.** Summing (12) and using $\Phi_{s_K}^K \ge \underline{\Phi}$ give

$$c_{\mathrm{desc}} \sum_{k=0}^{K-1} \rho_{s_k} D_k \le \Phi_{s_0}^0 - \underline{\Phi} + \sum_{k=0}^{\infty} e_k =: C_{\mathrm{desc}} < \infty.$$

Appendix II.1: Induction—Joint Claim

Contents

- **Summable drift budget.** Define

$$e_k := C_r^2 \left(\frac{\tau_k}{\rho_{s_k}^2} + \frac{\rho_{s_{k+1}} - \rho_{s_k}}{2\rho_{s_k}^2} \right), \qquad C_e := \sum_{k=0}^{\infty} e_k < \infty.$$

- **Derived level set.** Set

$$\overline{\Phi} := \Phi_{s_0}(\boldsymbol{x}^0, \boldsymbol{y}^0; \boldsymbol{\mu}^0) + C_e, \qquad \alpha := \overline{\Phi} + \frac{C_r^2}{2\rho_{s_0}}, \qquad \mathcal{S}_\alpha := \{\boldsymbol{x} : F(\boldsymbol{x}) \le \alpha\}.$$

  Choose a compact convex set $\mathcal{S}$ such that $\mathcal{S}_\alpha \subseteq \operatorname{int}(\mathcal{S})$.

- **Potential-to-objective bridge.** Multiplier boundedness and the Lipschitz continuity of h give

$$F(\boldsymbol{x}^k) \le \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) + \frac{C_r^2}{2\rho_{s_k}}.$$

- **Joint induction claim.** For every $k \ge 0$, prove simultaneously that

$$\Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) \le \Phi_{s_0}(\boldsymbol{x}^0, \boldsymbol{y}^0; \boldsymbol{\mu}^0) + \sum_{j=0}^{k-1} e_j, \qquad \boldsymbol{x}^k \in \mathcal{S}_\alpha.$$

- **Base case.** The potential bound is immediate, and

$$F(\boldsymbol{x}^0) \le \Phi_{s_0}(\boldsymbol{x}^0, \boldsymbol{y}^0; \boldsymbol{\mu}^0) + \frac{C_r^2}{2\rho_{s_0}} \le \alpha.$$

  Hence, $\boldsymbol{x}^0 \in \mathcal{S}_\alpha$.

Appendix II.2: Induction—Smooth Region

Contents

- **Induction hypothesis.** Assume that the joint claim holds at iteration k. In particular,

$$\boldsymbol{x}^k \in \mathcal{S}_\alpha \subseteq \operatorname{int}(\mathcal{S}).$$

- **Safety margin.** Define

$$d_{\mathcal{S}} := \operatorname{dist}(\mathcal{S}_\alpha, \partial\mathcal{S}) > 0, \qquad M_d := \max\left\{ \frac{\|\nabla_{\boldsymbol{x}} \Phi_{s_0}(\boldsymbol{x}^0, \boldsymbol{y}^0; \boldsymbol{\mu}^0)\|}{\rho_{s_0}}, \frac{G_f + G_{\mathcal{A}}(M_\mu + C_r)}{\rho_{s_0}} \right\}.$$

- **Normalized gradient bound.** Lemma 1 and Assumption (ii) imply

$$\frac{\|\nabla_{\boldsymbol{x}} \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k)\|}{\rho_{s_k}} \leq M_d.$$

  The definition of M_d covers $k = 0$; the bounds $\|\boldsymbol{\mu}^k\| \leq M_\mu$ and $\|\boldsymbol{r}^k\| \leq C_r/\rho_{s_{k-1}}$ cover $k \geq 1$.

- **The next iterate remains in $\mathcal{S}$.** Choose $\eta_k = \theta_x/\rho_{s_k}$ with $\theta_x M_d \leq d_{\mathcal{S}}$. Then

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^k\| = \frac{\theta_x}{\rho_{s_k}} \|\nabla_{\boldsymbol{x}} \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k)\| \leq \theta_x M_d \leq d_{\mathcal{S}}.$$

  Thus, $\boldsymbol{x}^{k+1} \in \mathcal{S}$. Convexity also gives $[\boldsymbol{x}^k, \boldsymbol{x}^{k+1}] \subseteq \mathcal{S}$.

- **Primal descent is now valid.** The whole line segment lies in the smoothness region. Therefore, the one-step descent inequality of Lemma 2 applies at iteration k.

Appendix II.3: Induction—Closure

Contents

- **Apply quasi-descent.** Since $\boldsymbol{x}^k, \boldsymbol{x}^{k+1} \in \mathcal{S}$, Lemma 2 and the drift bound give

$$\begin{aligned}\Phi_{s_{k+1}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) &\le \Phi_{s_k}(\boldsymbol{x}^k, \boldsymbol{y}^k; \boldsymbol{\mu}^k) + e_k \\ &\le \Phi_{s_0}(\boldsymbol{x}^0, \boldsymbol{y}^0; \boldsymbol{\mu}^0) + \sum_{j=0}^{k} e_j .\end{aligned}$$

- **Convert potential control to objective control.** Using $\sum_{j=0}^{k} e_j \le C_e$ and $\rho_{s_{k+1}} \ge \rho_{s_0}$,

$$\begin{aligned}F(\boldsymbol{x}^{k+1}) &\le \Phi_{s_{k+1}}(\boldsymbol{x}^{k+1}, \boldsymbol{y}^{k+1}; \boldsymbol{\mu}^{k+1}) + \frac{C_r^2}{2\rho_{s_{k+1}}} \\ &\le \overline{\Phi} + \frac{C_r^2}{2\rho_{s_0}} = \alpha .\end{aligned}$$

- **Return to the derived level set.** The definition of $\mathcal{S}_\alpha$ gives

$$\boldsymbol{x}^{k+1} \in \mathcal{S}_\alpha .$$

Both parts of the joint claim now hold at iteration $k+1$.

mechanism

$$\boldsymbol{x}^k \in \mathcal{S}_\alpha \implies \boldsymbol{x}^{k+1} \in \mathcal{S} \implies \text{quasi-descent is valid} \implies F(\boldsymbol{x}^{k+1}) \le \alpha \implies \boldsymbol{x}^{k+1} \in \mathcal{S}_\alpha .$$

Appendix III: Danskin's Theorem

Contents

Let

$$V(\mathbf{u}) := \max_{\mathbf{v}\in\mathcal{V}} g(\mathbf{u},\mathbf{v}), \qquad \mathbf{v}^\star(\mathbf{u}) \in \arg\max_{\mathbf{v}\in\mathcal{V}} g(\mathbf{u},\mathbf{v}).$$

Assume that the maximum is attained, g is continuously differentiable in $\mathbf{u}$, and the standard continuity conditions of Danskin's theorem [3] hold.

Unique-maximizer form

If $\mathbf{v}^\star(\mathbf{u})$ is unique, then V is differentiable at $\mathbf{u}$ and

$$\nabla V(\mathbf{u}) = \nabla_{\mathbf{u}} g\big(\mathbf{u}, \mathbf{v}^\star(\mathbf{u})\big).$$

No derivative of the solution map $\mathbf{v}^\star(\mathbf{u})$ is required.

Appendix IV: Why $\rho + s_k$ Appears

Contents

For fixed $(\boldsymbol{x}^{k+1}, \boldsymbol{\mu}^k)$, the two conditions are

$$0 \in \partial h(\boldsymbol{y}^{k+1}) - \boldsymbol{\lambda}^{k+1} + \rho\big(\boldsymbol{y}^{k+1} - \mathcal{A}(\boldsymbol{x}^{k+1})\big),$$

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\mu}^k + s_k\big(\mathcal{A}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}^{k+1}\big).$$

Substituting the second condition into the first gives

$$0 \in \partial h(\boldsymbol{y}^{k+1}) - \boldsymbol{\mu}^k + (\rho + s_k)\big(\boldsymbol{y}^{k+1} - \mathcal{A}(\boldsymbol{x}^{k+1})\big),$$

and therefore the explicit $(\rho + s_k)$-prox update follows.

Appendix V: Danskin's Theorem for the Max–Min Problem

Contents

For fixed $(\boldsymbol{x}, \boldsymbol{\mu})$, define the reduced dual function

$$d_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{\mu}) := \min_{\boldsymbol{y}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}).$$

The inner problem is ρ-strongly convex and has the unique minimizer

$$\boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}) = \operatorname{prox}_{h/\rho}\left(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\lambda}}{\rho}\right).$$

Thus, the max–min problem is equivalently

$$\max_{\boldsymbol{\lambda}} \min_{\boldsymbol{y}} \mathcal{P}_s(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}; \boldsymbol{\mu}) = \max_{\boldsymbol{\lambda}} d_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{\mu}).$$

Danskin's theorem

Since the inner minimizer is unique and $\mathcal{P}_s$ is differentiable with respect to $(\boldsymbol{x}, \boldsymbol{\lambda})$, Danskin's theorem gives

$$\begin{aligned}\nabla_{\boldsymbol{\lambda}} d_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{\mu}) &= \nabla_{\boldsymbol{\lambda}} \mathcal{P}_s\big(\boldsymbol{x}, \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}), \boldsymbol{\lambda}; \boldsymbol{\mu}\big) \\ &= \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}) - \frac{1}{s}(\boldsymbol{\lambda} - \boldsymbol{\mu}).\end{aligned}$$

Moreover,

$$\begin{aligned}\nabla_{\boldsymbol{x}} d_s(\boldsymbol{x}, \boldsymbol{\lambda}; \boldsymbol{\mu}) &= \nabla_{\boldsymbol{x}} \mathcal{P}_s\big(\boldsymbol{x}, \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x}), \boldsymbol{\lambda}; \boldsymbol{\mu}\big) \\ &= \nabla f(\boldsymbol{x}) + J_{\mathcal{A}}(\boldsymbol{x})^\top \left[\boldsymbol{\lambda} + \rho\big(\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x})\big)\right].\end{aligned}$$

No derivative of the possibly nonsmooth solution map $\boldsymbol{y}(\boldsymbol{\lambda}; \boldsymbol{x})$ with respect to either $\boldsymbol{x}$ or $\boldsymbol{\lambda}$ is needed.

Appendix VI: Exact Solution of the Max–Min Problem

Contents

The function $d_s(\boldsymbol{x}, \cdot; \boldsymbol{\mu})$ is $1/s$-strongly concave. Its unique maximizer $\boldsymbol{\lambda}^\star$ therefore satisfies

$$0 = \nabla_{\boldsymbol{\lambda}} d_s(\boldsymbol{x}, \boldsymbol{\lambda}^\star; \boldsymbol{\mu}) = \mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}(\boldsymbol{\lambda}^\star; \boldsymbol{x}) - \frac{1}{s}(\boldsymbol{\lambda}^\star - \boldsymbol{\mu}).$$

Let $\boldsymbol{y}^\star := \boldsymbol{y}(\boldsymbol{\lambda}^\star; \boldsymbol{x})$. The exact optimality conditions are

$$0 \in \partial h(\boldsymbol{y}^\star) - \boldsymbol{\lambda}^\star + \rho\big(\boldsymbol{y}^\star - \mathcal{A}(\boldsymbol{x})\big),$$

$$\boldsymbol{\lambda}^\star = \boldsymbol{\mu} + s\big(\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}^\star\big).$$

Substitution gives

$$0 \in \partial h(\boldsymbol{y}^\star) - \boldsymbol{\mu} + (\rho + s)\big(\boldsymbol{y}^\star - \mathcal{A}(\boldsymbol{x})\big).$$

Hence the unique exact solution is

$$\boxed{\boldsymbol{y}^\star = \operatorname{prox}_{h/(\rho+s)}\left(\mathcal{A}(\boldsymbol{x}) + \frac{\boldsymbol{\mu}}{\rho + s}\right), \qquad \boldsymbol{\lambda}^\star = \boldsymbol{\mu} + s\big(\mathcal{A}(\boldsymbol{x}) - \boldsymbol{y}^\star\big).}$$